

LabGuardian: Geometry-Constrained Pin Pose Estimation for Visual Breadboard Reconstruction

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Abstract—Recovering breadboard wiring from a single photograph requires terminal-level localization: holes are separated by a 2.54 mm pitch, leads are thin and often occluded, and a one-cell error changes the inferred circuit. Conventional component detectors terminate at bounding boxes, whereas unconstrained image-to-text models can generate connections unsupported by the pixels. We present LabGuardian, a geometry-aware visual reconstruction pipeline that casts terminal localization as component-conditioned pose estimation. A full-frame network jointly predicts component instances and ordered terminal keypoints; a planar homography then rectifies the board, and a package-aware snap-to-grid algorithm maps terminals to discrete holes while retaining ranked alternatives for ambiguous observations. The resulting pin-hole graph preserves links to the originating box, keypoint, rectified coordinate, and candidate holes. On a disjoint validation split, component detection achieves 0.991 mAP50 and 0.786 mAP50-95, while pin pose estimation achieves 0.947 and 0.829, respectively. INT8 inference on the neural processing unit of an Intel Core Ultra 5 225U requires 13.37 ms on average and 15.61 ms at the 99th percentile, delivers 74.7 images/s, and operates at 8.53 W package power. These results demonstrate accurate pin-level perception with interactive edge deployment.

Index Terms—pose estimation, keypoint localization, homography, visual circuit reconstruction, edge inference

I. INTRODUCTION

Converting a photograph of a physical circuit into an explicit structural description is a prerequisite for automated inspection of hand-wired hardware, for robotic assembly verification, and for assistive systems that check a prototype against an intended design. Breadboards make this problem concrete and unusually strict. Holes are arranged on a regular 2.54 mm lattice, and the electrical meaning of a component is determined entirely by which holes its terminals occupy. Two reconstructions that differ by a single grid cell describe different circuits: one may be the intended amplifier, the other an open loop or an unintended short. Accuracy therefore has to be defined at the level of individual holes rather than of component bodies.

Most visual pipelines stop before that level. A detector that returns a box labeled *resistor* identifies the presence of a device but not where its leads enter the board, and cropping the box before regressing terminals removes exactly the evidence

needed: axial leads extend well beyond the component body, and the surrounding lattice provides the reference against which a terminal position becomes meaningful. The natural representation is an instance-conditioned set of terminal keypoints predicted in the full image.

A second difficulty is geometric rather than semantic. Nearest-neighbor matching in image coordinates is not invariant to viewpoint: under an oblique camera the row spacing compresses, and a terminal can be attracted to the wrong row even when its pixel localization is correct. Purely learned image-to-topology models, in turn, have no explicit mechanism to enforce the planar lattice, the separation across the central trench, the span of an axial device, or the pin order of a dual in-line package. Multimodal language models provide flexible visual semantics [1], but a free-form description cannot guarantee that a claimed connection corresponds to observed pixels. We therefore treat unsupported topology generation as a distinct failure mode: in a verification setting, an unsupported claim is more difficult to audit than a missed detection because it carries no deterministic pixel-to-hole correspondence.

LabGuardian therefore decomposes reconstruction into visual estimation and geometric projection steps, each producing an inspectable intermediate result. A full-frame pose network predicts component instances and ordered terminal keypoints; a planar homography maps the board to a canonical metric plane; and a geometry-constrained snap-to-grid stage converts rectified coordinates into discrete hole hypotheses, keeping ranked alternatives wherever the observation does not support a unique assignment. Electrical interpretation and symbolic comparison consume this pin-hole graph but never modify it, so a structural conclusion can always be traced back to a bounding box, a keypoint, a rectified coordinate and a candidate hole.

The contributions of this paper are as follows.

- We formulate terminal localization as component-conditioned pose estimation on the full board image, and associate pin hypotheses with component instances through a multi-cue, category-gated matching score with explicit abstention.

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- We introduce a projective-to-discrete geometric interface that couples homography-based rectification with a feasibility-constrained snap-to-grid assignment, and that represents residual hole-level ambiguity explicitly instead of resolving it silently.
- We report accuracy and heterogeneous edge inference for the complete visual stage: 0.947 pin mAP50 and 0.829 pin mAP50-95, and 13.37 ms mean latency at 74.7 images per second within 8.53 W on an integrated neural processing unit.

II. RELATED WORK

A. Keypoint and Pose Estimation

Pose estimation predicts structured landmarks rather than only a category or box. HRNet preserves high-resolution features [2], whereas YOLO-Pose regresses boxes and keypoints jointly with instance grouping in one pass [3], [4]. We reinterpret this heatmap-free formulation for rigid components whose terminal landmarks are small, often collinear, and constrained by both package geometry and the receiving lattice.

B. Planar Rectification and Metric Grids

A planar homography estimated from four correspondences normalizes viewpoint [5]. Breadboards provide an approximately planar metric lattice, but rectification alone cannot resolve terminals near a hole boundary. We therefore couple it to discrete assignment constraints for package type, terminal span, collinearity, and trench separation, while retaining the reason for each rejected candidate.

C. Circuit Understanding and Symbolic Verification

Circuit datasets largely address schematic recognition, board inspection, or already symbolic netlists [6]; graph matching likewise assumes that a graph is available [7]. Generic multimodal models can summarize images but do not provide deterministic pixel-to-hole correspondence. We address this missing interface by recovering terminals and canonical board locations from a photograph. Symbolic comparison remains only a downstream consumer of the visual result.

III. METHOD

A. Problem Formulation

Let an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ show component instances $\mathcal{C} = \{C_i\}$. Each instance carries a class t_i , a bounding box b_i , and an ordered set of terminal keypoints $K_i = \{\mathbf{p}_{ik}\}$ with visibility flags. A board model supplies canonical hole centers $\mathcal{H} = \{\mathbf{c}_h\}$ together with feasibility relations induced by the board layout and by component packaging. The task is to estimate

$$F : I \longrightarrow \{(C_i, k, h, z_{ik})\}, \quad (1)$$

where h is the assigned hole and z_{ik} records keypoint confidence, rectified projection distance, candidate rank, visibility and an ambiguity flag. Throughout, a *pin* is the indexed terminal of a component or package, and a *terminal keypoint* is the image location at which that pin meets the board. The

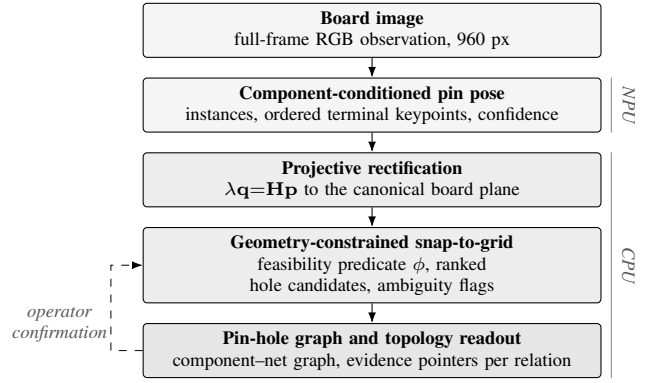


Fig. 1. Visual reconstruction pipeline. Perception, rectification and discrete assignment are separated so that every stage can be inspected independently. Topology readout consumes the pin-hole graph but does not alter the visual observations; a confirmed hole assignment re-runs only the geometric and symbolic stages. Vertical rules on the right indicate the execution unit that hosts each stage on the target platform.

formulation deliberately precedes electrical interpretation: it states what the image supports, not whether the circuit is correct. Fig. 1 summarizes the resulting stages.

B. Component-Conditioned Pin Pose Estimation

The pose network processes the complete image at 960 px resolution and, for every detected component, regresses a box, a class score, terminal coordinates, per-keypoint visibility and per-keypoint confidence. Full-frame inference is a deliberate choice: it preserves leads that leave the component body and the lattice that surrounds them, both of which a crop would discard. Seven classes are annotated directly: resistor, ceramic capacitor, electrolytic capacitor, diode, light-emitting diode, jumper wire and three-terminal transistor. Dense dual in-line packages are treated differently. Their pins occupy only a few pixels each and direct keypoint regression is unstable, so pin positions are derived from rigid package geometry once the package and its orientation have been detected.

The pose output jointly contains a component box and its ordered keypoints. Category-gated association is nevertheless required when these records are matched to annotations or fused with auxiliary component observations: a terminal assigned to a neighboring device propagates into an incorrect electrical relation. For component record i and pose record j , the implementation computes

$$S_{ij} = 1.8 \text{IoU}(b_i, b_j) + 0.45 \pi_{ij} + 0.90 \rho_{ij} + 0.65 \gamma_{ij} + 0.10 c_j, \quad (2)$$

where b is a bounding box, c_j is detection confidence, and $\pi_{ij} = \max(0, 1 - \|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|_2 / (1.5d_i))$ is center proximity normalized by the diagonal d_i of b_i . ρ_{ij} measures keypoint containment in a box dilated by 22% (with a 12-pixel minimum padding), and γ_{ij} measures agreement between the keypoint span and component extent. Incompatible classes are removed; pairs with $S_{ij} > 0.05$ are greedily assigned one to one in descending score order. This explicit rule makes instance ownership reproducible and prevents a terminal from being claimed twice.

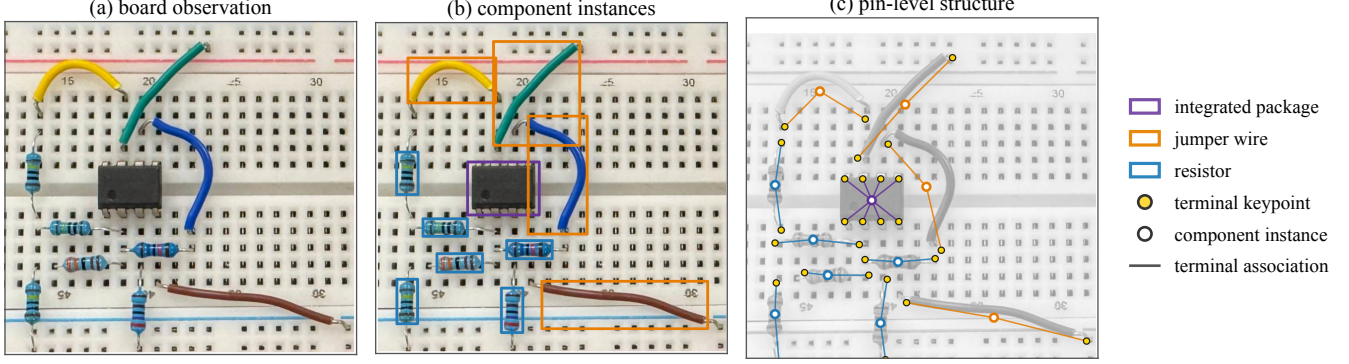


Fig. 2. Reconstruction on a real board. (a) full-frame observation; (b) component instances with class-coded boxes; (c) the instance-linked pin representation after geometric mapping, where each terminal keypoint is bound to a component node and to the hole it occupies. Panel (c) is the interface consumed by topology readout; it adds no visual evidence of its own.

C. Projective Rectification

Let $\mathbf{p} = [x_p, y_p, 1]^\top$ be a terminal keypoint in image coordinates. Four board landmarks determine a nonsingular planar homography $\mathbf{H} \in \mathbb{R}^{3 \times 3}$, and the canonical coordinate $\mathbf{q} = [u, v, 1]^\top$ follows from

$$\lambda \mathbf{q} = \mathbf{H} \mathbf{p},$$

$$u = \frac{h_{11}x_p + h_{12}y_p + h_{13}}{h_{31}x_p + h_{32}y_p + h_{33}}, \quad v = \frac{h_{21}x_p + h_{22}y_p + h_{23}}{h_{31}x_p + h_{32}y_p + h_{33}} \quad (3)$$

with $\lambda \neq 0$ the homogeneous scale and h_{ij} the entries of $\mathbf{H}$. Equation (3) removes projective distortion before any discrete decision is taken, which makes the acceptance radius of the next stage a physical quantity instead of a viewpoint-dependent one. When landmark calibration fails, a synthetic lattice is used as a fallback, but every assignment produced under the fallback is marked so that it is not treated as equally reliable evidence.

D. Geometry-Constrained Snap-to-Grid

Given the rectified coordinate and the canonical hole set, the nearest candidate and its normalized distance are

$$h^* = \arg \min_{h \in \mathcal{H}} \|\mathbf{q}_{1:2} - \mathbf{c}_h\|_2, \quad a = \frac{d_{h^*}}{s_g}, \quad (4)$$

$$\text{accept} = (a \leq \tau) \wedge (\phi(h^*, t_p, K_i) = 1), \quad (5)$$

where d_{h^*} is the rectified snapping distance, s_g the physical grid pitch, τ a normalized ambiguity threshold, t_p the terminal or package type, and ϕ a feasibility predicate. The implementation retains the five nearest holes before applying ϕ . The predicate rejects assignments that leave the board region, cross the central trench where the two halves are electrically separate, violate the span of an axial device, break the collinearity of a three-terminal device, or contradict package-specific pin order. Normalizing the snapping distance by s_g in (4) makes τ independent of output resolution.

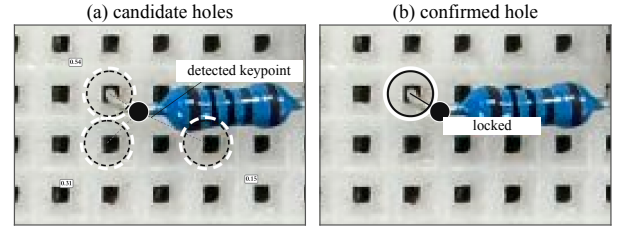


Fig. 3. Hole-level ambiguity. (a) A terminal that projects between adjacent holes is retained as several ranked candidates with their scores instead of being forced onto the nearest one. (b) After confirmation the terminal is bound to a single hole and the downstream graph is recomputed without re-running perception.

Rejection does not discard the observation. The assignment stage records the ranked candidate holes, their normalized distances, the violated constraint and a confidence flag, so a terminal that projects near the boundary between two holes is reported as ambiguous rather than resolved by an arbitrary tie-break (Fig. 3). This is a deliberate design decision. Dense millimeter-scale holes are an intrinsically hard localization target, so a residual hole-level error cannot be driven to zero by a stronger backbone alone, and a sub-pixel residual that is rounded silently becomes a false structural claim. Where a low-confidence assignment matters, a single operator confirmation resolves it and only the geometric and symbolic stages are recomputed; the perception result is not discarded.

E. Topology Readout

Accepted pin-hole relations are grouped by the known breadboard connectivity. A disjoint-set structure merges strips joined by jumper wires and produces a component-net graph in which nets are nodes and two-terminal components are edges; multi-pin packages retain numbered component nodes.

Because physical holes remain the source of net identity, correcting a hole regenerates rather than inherits net labels. The graph can be compared with a reference through subgraph isomorphism [7]. This readout adds no visual evidence and is not evaluated as a separate contribution; it only verifies that every reported relation remains traceable to its instance, keypoint, and candidate hole.

IV. EXPERIMENTS

A. Implementation and Data

We train and evaluate on a private dataset of real top-view breadboard photographs covering six textbook analog circuit topologies: a first-order RC network, a common-emitter amplifier, a differential pair, and inverting, integrating and summing amplifiers built around a general-purpose operational amplifier. The images were captured under varying viewpoint and illumination and include wire overlap and partial occlusion, so no synthetic augmentation of the scene layout was required. Annotations combine component boxes with up to three ordered terminal keypoints per instance and per-keypoint visibility. Training and validation splits are disjoint, and all metrics reported below are computed on the validation split excluded from optimization.

The pose model is YOLOv8s-pose [4], trained for 100 epochs at 960 px input with batch size 8, standard color, translation and scale augmentation, and mosaic augmentation disabled for the final ten epochs. Post-training INT8 quantization used 144 calibration images and reduced the exported model from 22 MB to 11 MB. All deployment measurements were taken on an edge platform built around an Intel Core Ultra 5 225U with 8 GB of memory, which exposes CPU, integrated GPU and neural processing unit execution paths through a common runtime [8]. Between 60 and 1153 sequential inferences were collected per configuration, and package power was sampled from the running average power limit interface at 0.25 s intervals.

B. Detection and Pin Pose Accuracy

Table I reports the Ultralytics validation outputs after 100 epochs. The keypoint head keeps precision and recall above 0.95 despite the small spatial support of a terminal. The gap between pin mAP50 and mAP50-95 is the more informative number for our purpose: coarse keypoint detection is largely solved, while strict localization is not, and it is precisely the strict regime that determines whether a terminal falls on the correct side of a hole boundary. This is the empirical motivation for treating rectification and constrained assignment as separate, inspectable stages rather than expecting the pose head to resolve hole identity by itself.

C. Edge Inference Performance

Table II lists latency and throughput for the six device-precision configurations. INT8 execution on the neural processing unit attains the lowest mean and tail latency, reducing mean latency by 85.5% relative to CPU FP16 and raising throughput from 60.4 to 74.7 images per second relative to

TABLE I
VALIDATION ACCURACY AFTER 100 EPOCHS

Prediction target	Precision	Recall	mAP50	mAP50-95
Component boxes	0.991	0.989	0.991	0.786
Pin keypoints	0.955	0.954	0.947	0.829

TABLE II
LATENCY AND THROUGHPUT OF THE POSE MODEL ACROSS EXECUTION UNITS

Device	Prec.	Load (s)	Mean (ms)	P95 (ms)	P99 (ms)	img/s
CPU	FP16	0.14	92.44	96.74	98.92	10.8
CPU	INT8	0.24	29.02	30.13	30.65	34.5
iGPU	FP16	0.43	26.87	27.32	28.18	37.2
iGPU	INT8	2.85	18.26	19.29	20.11	54.7
NPU	FP16	1.01	16.55	16.64	17.67	60.4
NPU	INT8	1.20	13.37	13.75	15.61	74.7

FP16 on the same unit. The 15.61 ms 99th-percentile latency is the relevant figure for interactive use, since it bounds the delay of the visual stage rather than its average cost.

Package-power measurements collected in the same session show that NPU INT8 execution reaches 8.53 W, approximately 4.1 W above the 4.4 W idle baseline, which corresponds to 114.2 mJ per inference. By comparison, CPU INT8 reaches 26.4 W at 34.5 images/s and consumes approximately seven times more energy per inference. During NPU execution, the CPU-core and integrated-GPU traces remain near their idle levels; these units therefore remain available for rectification, graph construction, and visualization.

D. Qualitative Analysis

Fig. 2 shows a complete reconstruction. Boxes and classes are recovered on a cluttered board, and the pin-level panel makes explicit which terminal belongs to which instance and which hole each terminal occupies. The representation also separates two distinct failure sources. Component boxes can be correct while terminal endpoints are partially hidden by jumper wires, and full-frame inference is what preserves the visible lead direction and neighboring lattice needed to place such endpoints. Independently, a terminal can be localized accurately and still fall near a boundary between two holes after rectification, as in Fig. 3; this is a geometric condition, not a perception error, and the pipeline reports it as such.

Fig. 4 shows the structure obtained on a second board, an inverting amplifier, in a single run of the complete pipeline: twelve components and 30 terminals resolve into ten nets, with supply and ground roles separated from signal nets and six terminals flagged as ambiguous rather than silently resolved. The example is included as evidence that the visual output is complete enough to be consumed symbolically, not as a claim about diagnostic accuracy. On this platform the deterministic path from a rectified pose result to a topology comparison completes in under 100 ms, so the visual stage dominates the end-to-end cost.

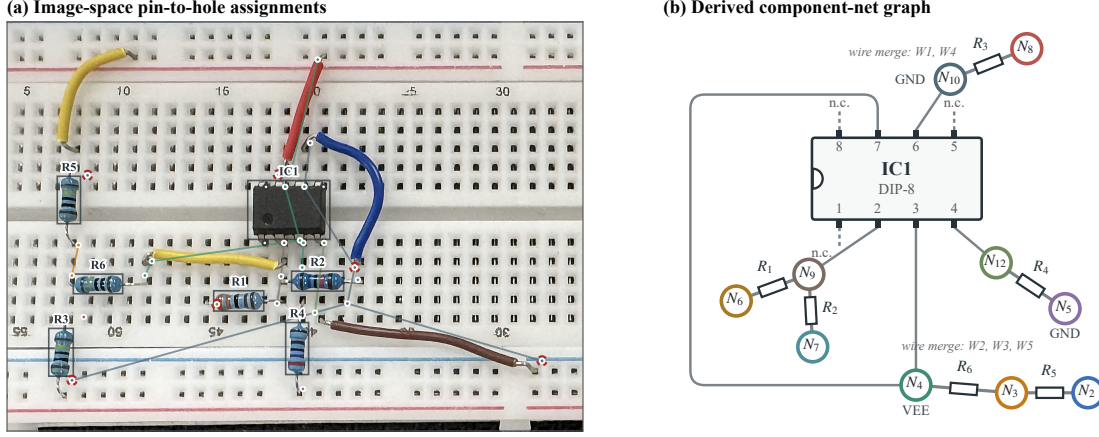


Fig. 4. Topology readout on a second board. (a) Terminals are bound to holes and colored by electrical net; dashed rings denote ambiguous assignments. (b) The exported component-net graph uses the same colors, providing a direct visual correspondence between image evidence and symbolic structure.

V. DISCUSSION AND LIMITATIONS

The reported metrics establish pin-pose accuracy and deployment cost, while the case studies verify that the geometric output can be consumed by topology readout. They do not yet isolate the contribution of rectification and constrained snapping. The current annotation contains image-space key-points but no per-terminal ground-truth hole identifier; consequently, pose average precision cannot be interpreted as hole-assignment accuracy. A complete geometric evaluation requires a held-out hole-labeled subset and should report normalized keypoint error and PCK before snapping, exact hole-assignment accuracy after snapping, and precision-coverage for ambiguity rejection. An ablation comparing image-space nearest-neighbor assignment, rectification alone, and the full feasibility-constrained mapper would then separate the effects of the three stages.

The private dataset uses rigid-terminal rather than articulated-joint keypoints, so it is not directly comparable with public pose benchmarks; releasing its annotations and board geometry is important for reproducibility. Moreover, symbolic inconsistency cannot certify the correctness of an individual keypoint. Visual and structural metrics must therefore remain separate, and ambiguous assignments must remain visible.

VI. CONCLUSION

We presented LabGuardian, which reframes breadboard understanding as pin-level pose estimation coupled to an explicit geometric interface. Full-frame, component-conditioned keypoint prediction preserves the lead and lattice context that cropping destroys; a planar homography maps arbitrary viewpoints to a canonical board; and a feasibility-constrained snap-to-grid stage converts rectified coordinates into discrete hole hypotheses while keeping residual ambiguity visible. The reconstruction is auditable because every downstream relation remains bound to the image evidence that produced it. Validation accuracy and cross-device measurements show

that the approach is accurate at the pose level and suitable for real-time execution on an integrated neural processing unit. A hole-labeled benchmark and controlled geometric ablation remain necessary to quantify the discrete mapping stage independently.

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